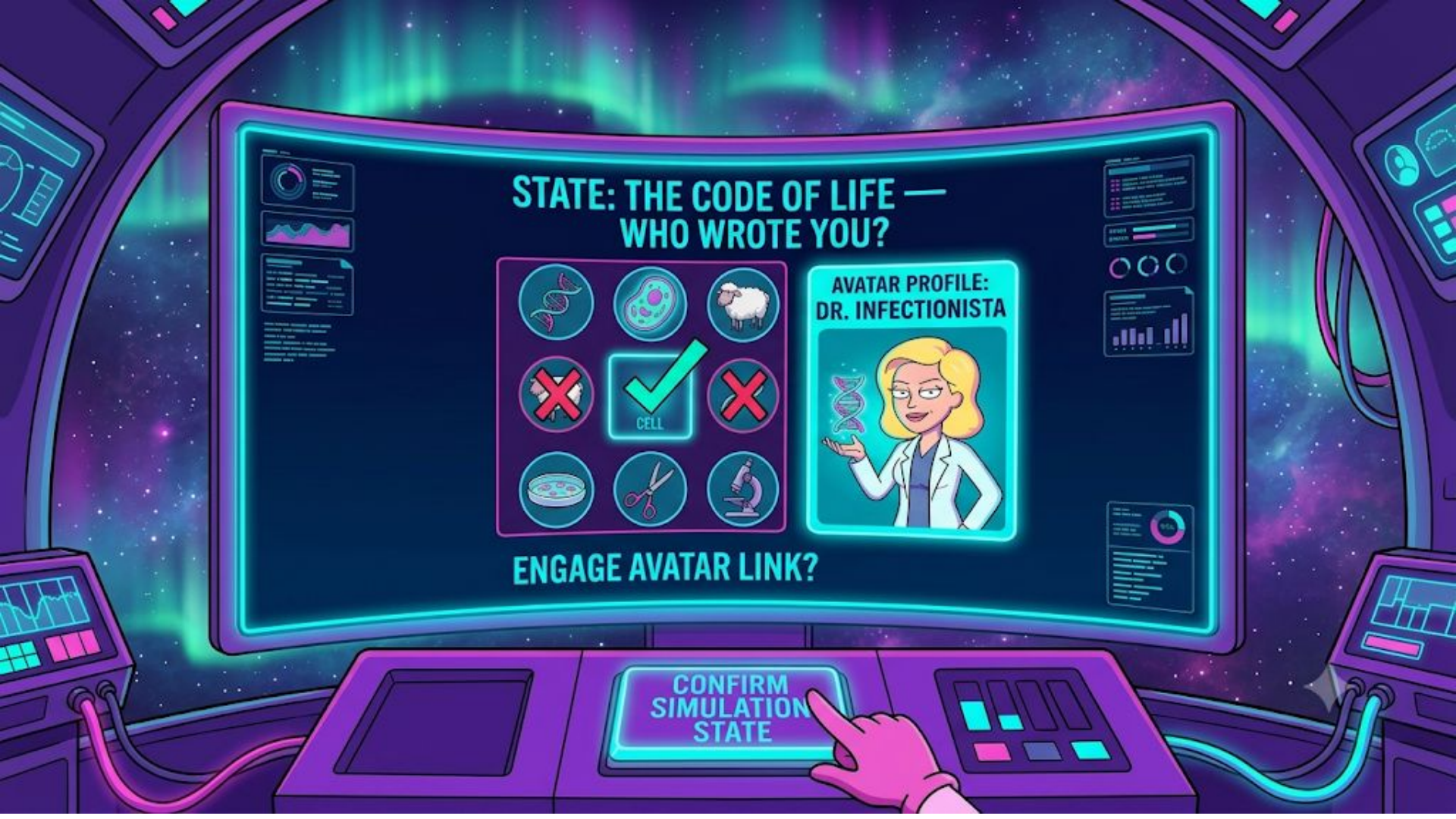

STATE: THE CODE OF LIFE — WHO WROTE YOU?
CELL
AVATAR PROFILE: DR. INFECTIONISTA
ENGAGE AVATAR LINK?
CONFIRM SIMULATION STATE

ONE CELL = ONE FACTORY



MEMBRANE — decides who gets in

security gates

NUCLEUS — the blueprints live here and never leave

MITOCHONDRIA — power plants with their OWN small blueprints

RIBOSOMES — print the proteins

LYSOSOME — recycling

stretch the DNA from ONE cell = **2 METRES** long

ALMOST EVERY CELL CARRIES THE WHOLE BLUEPRINT.

PLANT CELL vs ANIMAL CELL

PLANT CELL



- CELL WALL — a stiff box
- CHLOROPLASTS — solar panels
- BIG VACUOLE — water tank

ANIMAL CELL



- NO WALL — soft and squishy
- NO CHLOROPLASTS — it eats instead
- TINY VACUOLES

BOTH HAVE: NUCLEUS · MITOCHONDRIA · MEMBRANE · CYTOPLASM

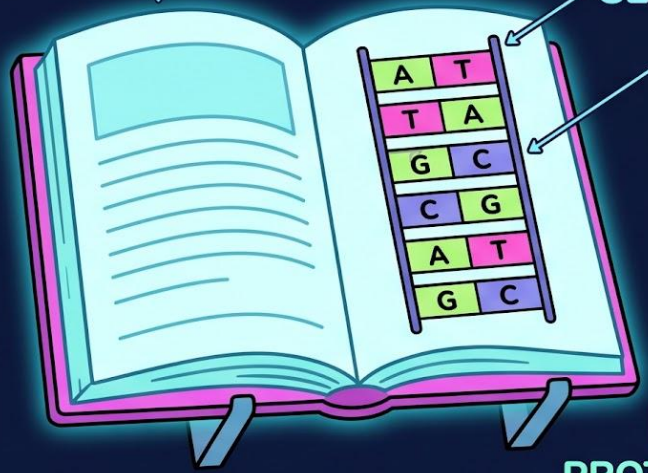
PLANTS BUILD THEIR OWN FOOD. ANIMALS GO AND GET IT.

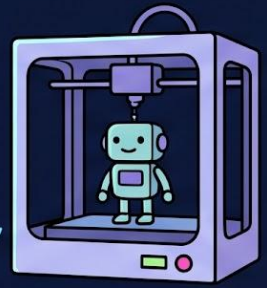
A GENE = ONE RECIPE

GENOME — the entire cookbook, ~20,000 recipes

GENE — a single recipe

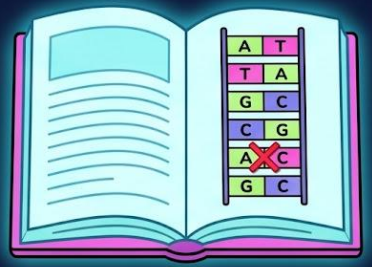
DNA — an alphabet of just 4 letters

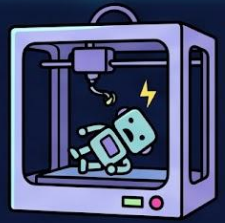




PROTEIN — the machine the recipe builds

GENE



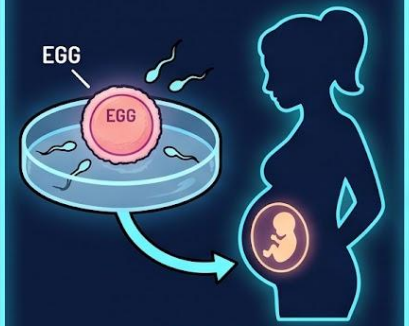


A TYPO → A BROKEN MACHINE → DISEASE

MOST TYPOS DO NOTHING. SOME BREAK A MACHINE.

THREE THINGS PEOPLE ALWAYS MIX UP

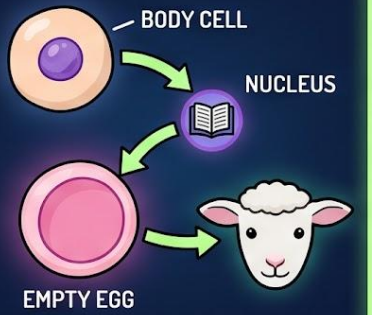
WHERE?



IVF

GENES: NOT CHANGED

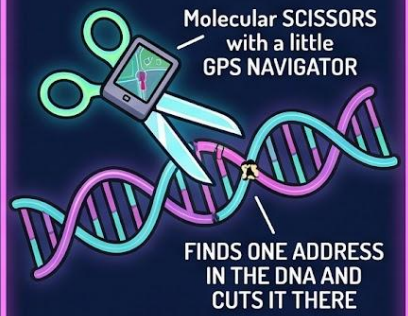
WHOSE?



CLONING

GENES: COPIED, NOT CHANGED

WHAT?



GENE EDITING

GENES: REWRITTEN

WHERE · WHOSE · WHAT — that's the whole difference.

IT ALREADY HAPPENED

STOP 1
1978 — THE FIRST DISH BABY



1

IVF — over 12 million children since

STOP 2
1996 — THE COPIED LAMB



2

cloned from an adult cell — 277 tries, 1 lamb

STOP 3
2018 — THE LINE WAS CROSSED



3

embryos were edited in secret — the rules were broken, the work was condemned

STOP 4
2023 — THE FIRST APPROVED FIX



4

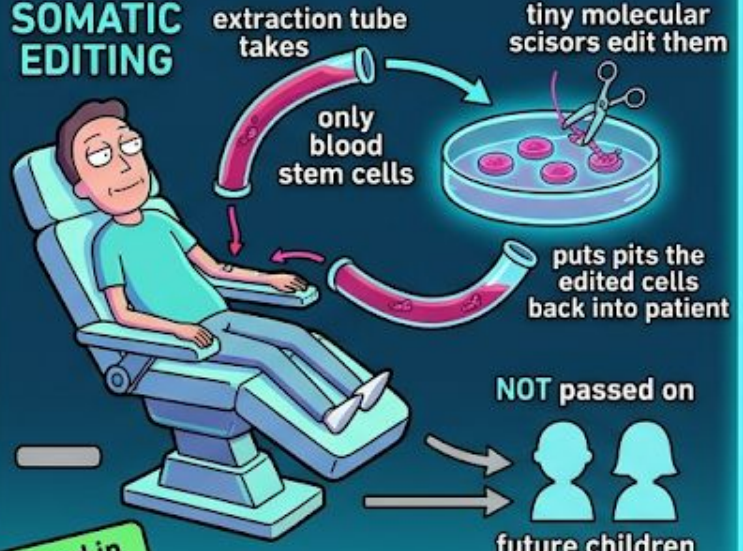
first approved CRISPR medicine — for SICKLE CELL disease

SAME TOOL. VERY DIFFERENT CHOICES.

WHERE'S THE LINE?

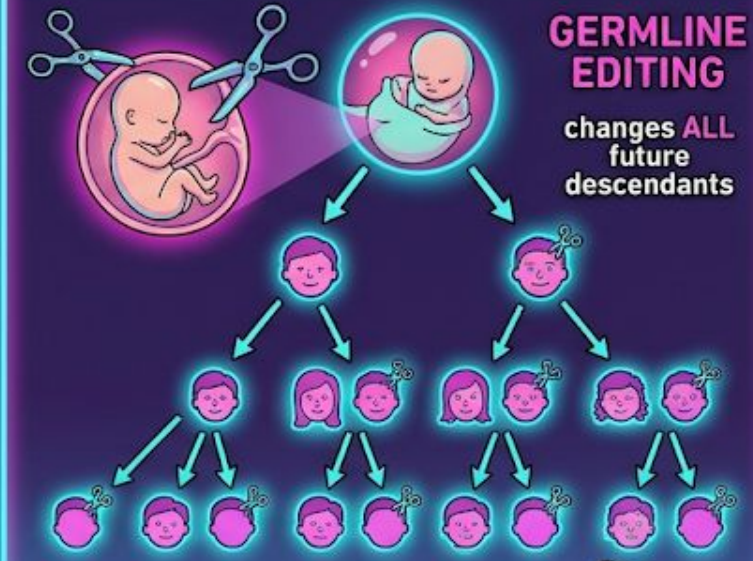
FIXING ONE PERSON

SOMATIC EDITING



allowed in medicine today

CHANGING EVERY GENERATION



GERMLINE EDITING

changes ALL future descendants

PASSED ON FOREVER

banned in most countries

WHO ASKS PERMISSION FROM PEOPLE NOT BORN YET?

HANDS UP: IS THIS OK?

1



Edit a 14-year-old's blood stem cells to treat sickle cell disease

VOTES:

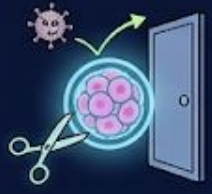
2



Edit an embryo to remove a gene that kills children before age 5

VOTES:

3



Edit an embryo to protect it from HIV — the 2018 case

VOTES:

4



Edit an embryo so the child grows taller

VOTES:

WHERE DID THIS ROOM DRAW THE LINE?

MENDEL'S RULE: TWO CARDS, ONE SHOWS

THE DEAL



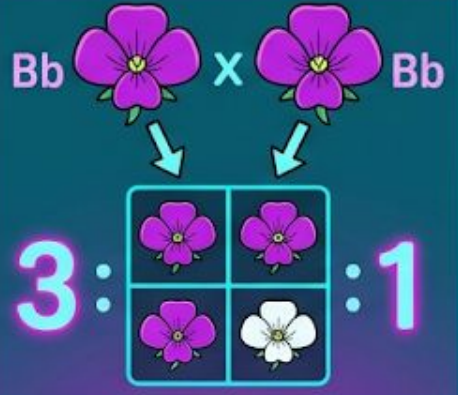
YOU GET ONE CARD FROM EACH PARENT

THE LOUD ONE WINS



ONE LOUD CARD IS ENOUGH TO SHOW

THE HIDDEN CARD



TWO PURPLE PARENTS CAN HAVE A WHITE CHILD

A CARD CAN HIDE FOR A WHOLE GENERATION — AND STILL BE DEALT.

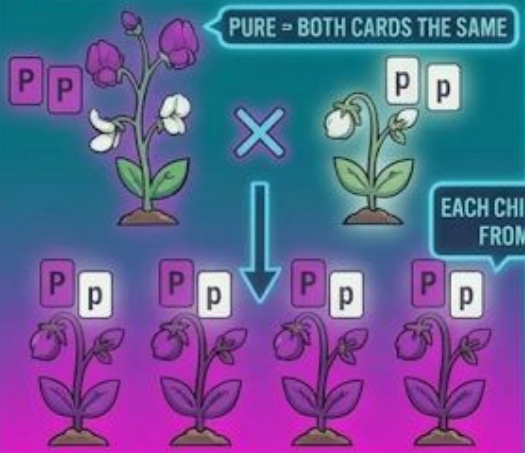


GREGOR MENDEL

- 1656-1663
- a monk in a monastery garden in Brno
- - 28,000 pea plants, pollinated by kand and COUNTED
- published 1866

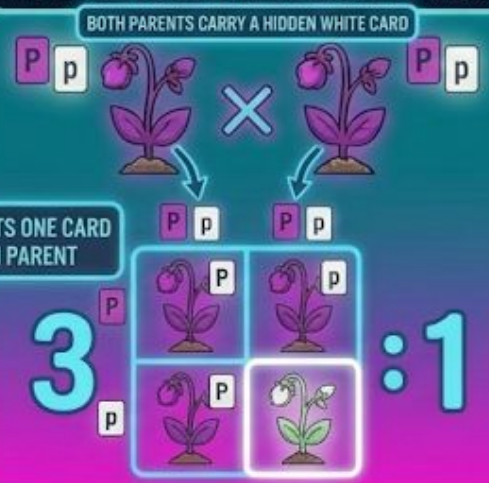
MENDEL'S THREE RULES

RULE 1 · WHY 'PURE' MATTERS



ALL PURPLE — BUT EVERY CHILD STILL CARRIES A WHITE CARD

RULE 2 · THE HIDDEN ONE RETURNS



ONLY TWO HIDDEN CARDS TOGETHER MAKE A WHITE FLOWER

RULE 3 · TRAITS TRAVEL ALONE



COLOUR AND SHAPE ARE DEALT SEPARATELY

HE COUNTED 28,000 PLANTS — AND NOBODY NOTICED FOR 34 YEARS.